AEGIS

A low-cost, modular, AI-assisted field network that helps small farmers save water and protect crops.

Techathon 3.0 · Vision Venture

Agri-Tech for Sustainable Farming

Team: [Your Name] · [Teammate]

[Your School]

FARMING RUNS BLIND

Small farmers guess — and pay for it in water and lost crops.

~80% of India's fresh water goes to agriculture — much of it wasted on over-irrigation.
₹ Lakhs Commercial smart-farming kits cost far beyond a small farmer's reach.
No alert Frost, heat, dryness and field fires are noticed only after damage is done.

Farmers water on a fixed schedule instead of on need, and get no early warning when conditions turn against the crop.

THE GAP

Existing systems are expensive, fragile and closed.

- **Too costly** — priced for large commercial farms, not a 2-acre holding.
- **Not repairable** — one failed part means replacing the whole unit.
- **Needs a smartphone + app** — excludes farmers on basic phones.
- **Dies offline** — no signal, no function, no warning.

THE IDEA

AEGIS puts a cheap, repairable sensor node in the field and the intelligence in the cloud.

Each node measures soil moisture, temperature, humidity and smoke. It sends readings over Wi-Fi to the AEGIS cloud, which decides what's happening and replies with plain-language advice — shown on the node's screen, on a live dashboard, and as an **SMS that works on any phone**.

HOW IT WORKS

Four layers. The node stays dumb; the cloud stays smart.

Field Node · ESP32
soil · temp · humidity · smoke

→

Wi-Fi · JSON
HTTP POST every 5s

→

Cloud API
Next.js / TypeScript

↳

Rule engine
instant status + risk

+

AI advice
Claude → plain language

→

Dashboard · SMS · LCD
farmer sees + acts

Because the node computes a safe local status too, the farmer **always** sees a warning on the screen — even with no internet. Resilience by design.

WORKING PROTOTYPE

Built, wired, live.

- ESP32 node on a breadboard, reading all four sensors.
- Live cloud dashboard with risk score, trends & alert history.
- Rule engine + AI advice returning real decisions.

AEGIS cloud dashboard — "IRRIGATE NOW", risk 62%, soil 28%.

FAST RULES + FRIENDLY AI

Decisions in milliseconds, explained like a human.

NODE SENDS →


```
{
  "soilPct": 22,
  "temperature": 34,
  "humidity": 38,
  "smoke": 700
}
```


← AEGIS REPLIES


```
{
  "status": "IRRIGATE NOW",
  "risk": 62,
  "message": "Soil low (22%).
   Irrigate field-a now; avoid
   over-watering to save water."
}
```


The rule engine guarantees safety and speed; the AI layer only **rephrases** the facts — so the system still works if the AI is ever offline.

SUSTAINABLE BY DESIGN

Every part of AEGIS pushes toward less waste.

Water

Irrigate on *need*, not schedule — directly cuts water use on every field.

Yield

Early frost / heat / fire / disease warnings prevent crop loss and wasted inputs.

E-waste

Modular, repairable nodes — replace a ₹-cheap sensor, not the whole device.

Accessible: SMS alerts reach any basic phone — no smartphone, no app, no data plan required.

WHAT MAKES IT WIN

Cheap. Modular. Unstoppable.

- **~1/100th the cost** of commercial kits — built from commodity parts.
- **Modular PCBs** — each sensor is its own replaceable board.
- **Works offline** — local status on the node when the cloud is down.
- **Any phone** — plain-language SMS, no app needed.
- **Scales** — add nodes across fields; one dashboard tracks them all.

ONE FIELD NODE

A full node for the price of a meal.

COMPONENT	ROLE	APPROX ₹
ESP32 DevKit	controller + Wi-Fi	350
Soil moisture sensor	irrigation signal	60
DHT11	temp + humidity	80
MQ-2	smoke / fire	90
16×2 I^2C LCD	on-site readout	150
Wiring, board, misc	—	120
Per node		**≈ ₹850**

Indicative retail prices; fall further at scale. Cloud runs on free/low-cost tiers.

WHERE IT GOES

From breadboard to the field.

Now ✓
node + cloud + AI + dashboard

→

Phase 2
SMS alerts (SIM/Twilio)

→

Phase 3
solar power · LoRa range

↳

Phase 4
auto irrigation valve

→

Phase 5
local-language voice + multi-farm

THANK YOU

Watch the field.
Save the water.

AEGIS — protection that any farmer can afford, repair, and understand.

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[contact / email]